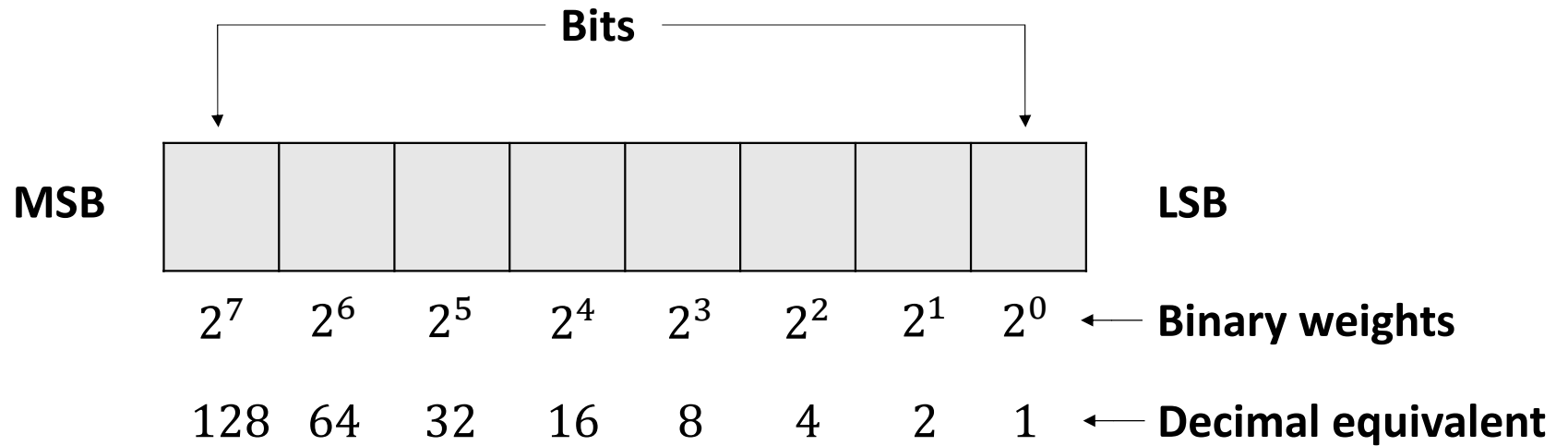


○ Bitwise Operators:

Bitwise operators are used to perform operation directly on the individual bits of integers. They treat integers as sequence of 1's and 0's and manipulate them on low-level.

Operators	Example	Description
& (Bitwise AND)	12 & 13	1100 & 1101 = 1100 (decimal 12)
(Bitwise OR)	12 13	1100 1101 = 1101 (decimal 13)
~ (Bitwise NOT)	~12	Equivalent to $-(x+1)$
^ (Bitwise XOR)	12 ^ 13	1100 ^ 1101 = 0001 (decimal 1)
<< (left shift)	5 << 2	0101 << 2 = 10100 (decimal 20)
>> (Right shift)	20 >> 2	10100 >> 2 = 00101 (decimal 5)

Binary Number System



Binary representation of 12

0	0	0	0	1	1	0	0	
0x16	0x64	0x32	0x16	1x8	1x4	0x2	0x1	
0	+	0	+	0	+	0	+	0
12								

Operators Precedence and Associativity:

Operator	Type	Precedence	Associativity
()	Parentheses	Highest	Left to right
**	Exponentiation	High	Right to left
* / % //	Multiplication, division, modulus, floor division	Medium	Left to right
+ -	Addition, subtraction	Medium	Left to right
< <= > >=	Comparison	Low	Left to right
== !=	Equality	Low	Left to right
not	Logical NOT	Low	Right to left
and	Logical AND	Lowest	Left to right
or	Logical OR	Lowest	Left to right

Operators Precedence and Associativity:

Sample Expression

Expression 1

$$\Rightarrow 10 + 3 * 2 ** 3 - (6 // 2 + 1) * 3 \% 4$$

$$\Rightarrow 10 + 3 * 2 ** 3 - (3 + 1) * 3 \% 4$$

$$\Rightarrow 10 + 3 * 2 ** 3 - 4 * 3 \% 4$$

$$\Rightarrow 10 + 3 * 8 - 4 * 3 \% 4$$

$$\Rightarrow 10 + 24 - 12 \% 4$$

$$\Rightarrow 10 + 24 - 0$$

$$\Rightarrow 34 - 0$$

$$\Rightarrow 34$$

Expression 2

$$\Rightarrow 10 / 2 + 3 * 2 ** 3 - (6 // 2 + 1) * 3 \% 4$$

$$\Rightarrow 10 / 2 + 3 * 2 ** 3 - (3 + 1) * 3 \% 4$$

$$\Rightarrow 10 / 2 + 3 * 2 ** 3 - 4 * 3 \% 4$$

$$\Rightarrow 10 / 2 + 3 * 8 - 4 * 3 \% 4$$

$$\Rightarrow 5 + 3 * 8 - 4 * 3 \% 4$$

$$\Rightarrow 5 + 24 - 4 * 3 \% 4$$

$$\Rightarrow 5 + 24 - 12 \% 4$$

$$\Rightarrow 5 + 24 - 0$$

$$\Rightarrow 29 - 0$$

$$\Rightarrow 29$$

Expression 3

$$\Rightarrow 10 * 2 + 3 * 2 ** 3 - (6 // 2 + 1) / 3 \% 4$$

$$\Rightarrow 10 * 2 + 3 * 2 ** 3 - (3 + 1) / 3 \% 4$$

$$\Rightarrow 10 * 2 + 3 * 2 ** 3 - 4 / 3 \% 4$$

$$\Rightarrow 10 * 2 + 3 * 8 - 4 / 3 \% 4$$

$$\Rightarrow 10 * 2 + 3 * 8 - 4 / 3 \% 4$$

$$\Rightarrow 20 + 3 * 8 - 4 / 3 \% 4$$

$$\Rightarrow 20 + 24 - 4 / 3 \% 4$$

$$\Rightarrow 20 + 24 - 1.3333 \% 4$$

$$\Rightarrow 20 + 24 - 1.3333$$

$$\Rightarrow 44 - 1.3333$$

$$\Rightarrow 42.66$$

Expressions:

```
>>> x = 10
>>> result = x * 5 + 2      # Arithmetic expression
>>> print(result)
52

>>>
>>> # Boolean expression
>>> age = 23
>>> is_adult = age >= 18
>>> print(is_adult)
True
>>> young_adult = age >= 18 and age <= 39
>>> print(young_adult)
True

>>>
>>> # String expression
>>> full_name = "John" + " " + "Doe"
>>> print(full_name)
John Doe

>>>
>>> # function call expression
>>> length = len("example")
>>> print(length)
7

>>>
>>> # literal expression
>>> number = 123
>>> print(number)
123
```

Basic Python Tasks:

1. Write Python program to take radius of circle as input and computes the area of circle ($A = \pi r^2$). Let $\pi = 3.14$
2. Write Python program to compute area of triangle ($\frac{1}{2} \times b \times h$)
3. Write Python program to solve $(x + y) \times (x + y)$. Let $x = 4$ and $y = 3$
4. Write Python program to find distance between two points (x_1, y_1) and (x_2, y_2) . Points coordinates are input by user.

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Control Flow:

○ Decision Making Structures: (If-else)

General structure:

```
if test1:  
    statement1  
elif test2:  
    statement2  
else:  
    statement3
```

```
a = int(input("Enter first number: "))
Enter first number: 8
b = int(input("Enter second number: "))
Enter second number: 6
c = int(input("Enter third number: "))
Enter third number: 4

if a>b and b>c:
    print("both conditions are true")
else:
    print("conditions are false")

both conditions are true
.
```